

NOTES 04: CATEGORICAL VARIABLES

Stat 120 | Spring 2026

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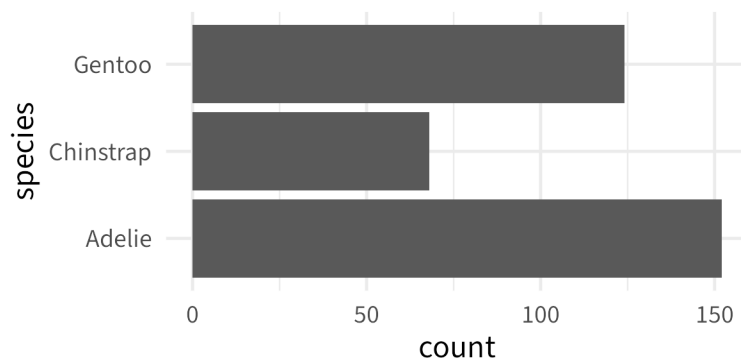
1 Categorical Variables

Categorical variables are best summarized with a **frequency table** and visualized using a **barplot**.

```
table(penguins$species)
```

Adelie	Chinstrap	Gentoo
152	68	124

```
ggplot(penguins, aes(y = species)) +  
  geom_bar()
```



If we want to summarize a categorical variable with a single number, or choose an appropriate parameter/statistic for a categorical variable, we often use a **proportion**.

Proportion

When we have two categorical variables, we often use a **two-way table** to summarize them at the same time (also called the **joint distribution**).

```
table(penguins$species, penguins$island)
```

	Biscoe	Dream	Torgersen
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

We might also care about the **marginal distribution** (the margins)

```
table(penguins$species, penguins$island) |>
  addmargins()
```

	Biscoe	Dream	Torgersen	Sum
Adelie	44	56	52	152
Chinstrap	0	68	0	68
Gentoo	124	0	0	124
Sum	168	124	52	344

or **conditional distribution** (a specific row/column).

```
table(penguins$species, penguins$island) |>
  proportions(2) # condition on COLUMNS
```

	Biscoe	Dream	Torgersen
Adelie	0.2619	0.4516	1.0000
Chinstrap	0.0000	0.5484	0.0000
Gentoo	0.7381	0.0000	0.0000

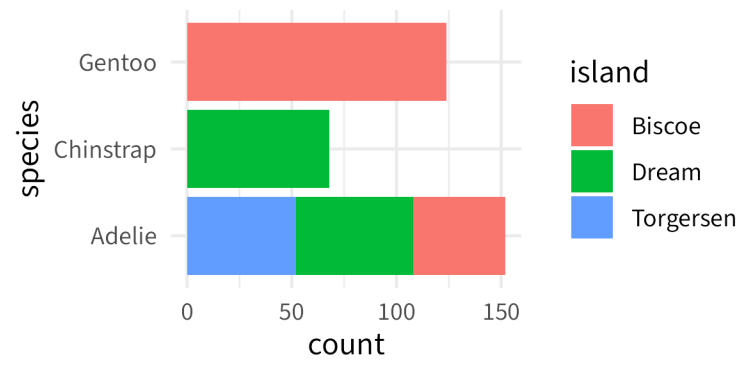
Independence

Two categorical variables are **independent** if the marginal and conditional distributions are the same. This means that knowing something about one variable tells us nothing about the distribution of the other variable.

Example: Is penguin species independent of island?

Visually, we use a filled barplot:

```
ggplot(penguins, aes(y = species, fill = island)) +
  geom_bar()
```



Example: Below is the two-way table for our class representing the answers to “What is your class year?” and “Coffee or tea?”

	First year	Sophomore
Coffee	10	8
Neither	3	0
Tea	5	5

- What is the marginal distribution of *coffee_or_tea*?
- What is the conditional distribution of *class year* among those who *prefer coffee*?
- What is the conditional distribution of *coffee or tea* among those who *sophomores*?
- Does *coffee or tea* appear to be independent of *class year*?